



Review

Automated seizure detection systems and their effectiveness for each type of seizure[☆]A. Ulate-Campos^{a,*}, F. Coughlin^b, M. Gaínza-Lein^{b,c}, I. Sánchez Fernández^{b,d}, P.L. Pearl^b, T. Loddenkemper^b^a Department of Neurology, National Children's Hospital "Dr. Carlos Saenz Herrera", San José, Costa Rica^b Division of Epilepsy and Clinical Neurophysiology, Department of Neurology, Boston Children's Hospital, Harvard Medical School, Boston, MA, USA^c Universidad Austral de Chile, Valdivia, Chile^d Department of Child Neurology, Hospital Sant Joan de Déu, Universitat de Barcelona, Spain

ARTICLE INFO

Article history:

Received 18 April 2016

Received in revised form 23 May 2016

Accepted 7 June 2016

Keywords:

Epilepsy

Seizure prediction sensors

Seizure alarms

Automated seizure detection

Intractable epilepsy

SUDEP

ABSTRACT

Epilepsy affects almost 1% of the population and most of the approximately 20–30% of patients with refractory epilepsy have one or more seizures per month. Seizure detection devices allow an objective assessment of seizure frequency and a treatment tailored to the individual patient. A rapid recognition and treatment of seizures through closed-loop systems could potentially decrease morbidity and mortality in epilepsy. However, no single detection device can detect all seizure types. Therefore, the choice of a seizure detection device should consider the patient-specific seizure semiologies.

This review of the literature evaluates seizure detection devices and their effectiveness for different seizure types. Our aim is to summarize current evidence, offer suggestions on how to select the most suitable seizure detection device for each patient and provide guidance to physicians, families and researchers when choosing or designing seizure detection devices. Further, this review will guide future prospective validation studies.

© 2016 Published by Elsevier Ltd on behalf of British Epilepsy Association.

1. Introduction

Epilepsy is one of the most common neurological disorders with an incidence of approximately 40–70/100,000 per year in adults [1] and 41–187/100,000 per year in children [2], being particularly frequent in rural and underdeveloped areas [1,3–5]. Almost 47% of patients will become seizure-free with the first anti-seizure medication trial and an additional 14% of patients with a second or third medicine [6]. Despite optimal medication management, about 20–30% of patients with epilepsy will continue to have more

than one seizure per month, 12% will have weekly seizures, and 8% daily seizures [3,7,8]. Patients with active epilepsy have 4–5 times higher standardized mortality ratios than epilepsy patients who are seizure free, especially in the initial years after a diagnosis of epilepsy [9–13]. The mortality associated with epilepsy has remained stable in the last 50 years despite the introduction of multiple new anti-seizure medications [14,15]. Further, seizure unpredictability worsens the quality of life (QOL) of patients with epilepsy and their families [16]. Medical treatment tailored to the individual patient's seizures might eventually decrease mortality and increase QOL, but in order to do so, both seizures and response to treatment need to be objectively quantified.

Currently, seizure tracking relies on subjective patient and family recall and may be influenced by the capacity to identify seizures, the level of awareness during the event, and the ability to remember details afterward. In a series of 451 children with absence seizures, up to 30% of patients with no clinically detected seizures, even during hyperventilation, had seizures detected on 1 h-EEG recordings [17,18]. In an inpatient study the number of seizures detected with EEG was 29 times higher than the number reported by families and 7 times higher than the number clinically observed by nurses [19]. In several studies patients report only

[☆] One of the authors of this paper is a member of the current editorial team of Seizure. The supervision of the independent peer review process was undertaken and the decision about the publication of this manuscript were made by other members of the editorial team.

* Corresponding author at: Department of Neurology, National Children's Hospital "Dr. Carlos Saenz Herrera", San José, Costa Rica. Tel.: +506 86674141.

E-mail addresses: adrianaulate@hotmail.com (A. Ulate-Campos),

francesca.coughlin@childrens.harvard.edu (F. Coughlin),

marina.gainzalein@childrens.harvard.edu (M. Gaínza-Lein),

ivan.fernandez@childrens.harvard.edu (I. Fernández),

phillip.pearl@childrens.harvard.edu (P.L. Pearl),

tobias.loddenkemper@childrens.harvard.edu (T. Loddenkemper).

Table 1

Types of seizure detection devices.

Electroencephalogram (EEG)
Intracranial EEG
Surface electromyography (sEMG)
Electrodermal activity (EDA)
Electrocardiography (EKG)
Accelerometry (ACM)
Video detection systems
Mattress sensors
Seizure-alert dogs
Implanted advisory systems
Cerebral oxygen saturation sensors
Near infrared spectroscopy (NIRS)
Skin temperature
Respiratory monitor

approximately half of their seizures, and even less during sleep [20–22]. Seizure detection devices provide more accurate seizure quantification, allowing clinicians to tailor treatment more objectively. In addition, seizure prediction devices may alert when an upcoming seizure is going to occur and may enhance patient and family confidence and improving QOL.

The main seizure detection modalities are outlined in Table 1. There is extensive literature describing seizure detectors, but limited data on which are optimal for each seizure type [23]. This manuscript aims to address this gap in knowledge by providing a guide on how to select specific devices for individual seizure types.

2. Seizure types

2.1. Seizure type definitions

Hughlings Jackson defined an epileptic seizure in 1870 as a “symptom ... of an occasional, an excessive and a disorderly discharge of nerve tissue” [24]. The definition of epileptic seizure was reformulated by the ILAE group in 2005 to, “a transient occurrence of signs and/or symptoms due to abnormal excessive or

synchronous neuronal activity in the brain” [24]. Specific seizure types have varying features that may enable detection (Table 1).

2.2. Different seizures are best captured by different types of sensors

Knowing the main semiological components of seizures is a basic initial step in the process of selecting the best seizure detection device for each patient. Each seizure type consists of one or more phenomena occurring simultaneously or sequentially [25]. In order to evaluate clinical features, the two main components that can be assessed are movement and physiological signals. The movement refers to specific body parts, as the limbs involved in a GTCS, or head or eye deviation. These could be detected by accelerometry, surface electromyography (sEMG), video monitoring, mattress sensors, electro-oculogram, or seizure-alert dogs. The physiological signals include heart rate, respiratory rate, sweating and temperature. These could be detected by electrocardiogram (EKG), sweating by EDA, temperature by a wristband and changes in respiratory rate with a thoracic band. Generalized tonic-clonic seizures (GTCS) may present with violent body movements and often prominent autonomic changes. Therefore, several sensor types recognize GTCS more readily than other seizure types. On the other end of the spectrum, absence are challenging to capture as they consist of a brief decrease in awareness with minimal associated movements (Table 2), and thus are often only picked up by an observer or by EEG.

The ideal seizure detection sensor should be able to detect movement in body parts and changes in physiological signals simultaneously. The system could be even more effective if it also interacted with the patient in an active way that could allow testing for awareness. This could be done with a gaming device that keeps track of the patient's responsiveness while playing. If the patient stops playing, the device could ask the patient to follow some specific commands, and in turn the system would ideally alert a designated caregiver in the event of impaired performance. This could also be done by robot or external device that wirelessly receives the input of movement and physiological signals from the monitoring device. For example a wristband analyzes signals in

Table 2

Definitions and main findings of the principal seizure types.

Seizure type	Main features Definition	Movement	Sweating	HR/EKG changes
Atonic	Sudden loss of muscle tone lasting 1–2 s, involving head, trunk, jaw or limb musculature	±	*	*
Autonomic	An alteration of the autonomic nervous system, involving cardiovascular, pupillary, gastrointestinal, sudomotor, vasomotor and thermoregulatory functions	—	+	+
Clonic	Semirhythmic high-amplitude movements that involve the same muscle groups	+	+	+
Myoclonic	Sudden, brief involuntary single or multiple low-amplitude contraction(s) of muscle(s) or muscle groups	+	*	—
Epileptic spasm	Sudden flexion and/or extension of predominantly proximal muscles, more sustained than a myoclonic movement but not so prolonged as a tonic seizure	+	*	*
Focal dyscognitive seizure	Disturbance of cognition is the most apparent feature accompanied by changes in perception, attention, emotion, memory and executive function	±	+	+
GTCS	Bilateral symmetric tonic contraction followed by generalized clonic movements of somatic muscles, usually accompanied by autonomic phenomena	+	+	+
Hypermotor	Involves predominantly proximal limb or axial muscles producing irregular, sequential semipurposeful movements.	+	+	+
Tonic	A sustained increase in muscle contraction lasting seconds to minutes	+	+	+

*No data on this particular issue.

Source: Blume, W.T et al. Glossary of descriptive terminology for ictal semiology: report of the ILAE task force on classification and terminology. *Epilepsia*, 2001; 42(9): p. 1212–8.

Legend: EKG: Electrocardiogram. GTCS: Generalized tonic-clonic seizure. HR: Heart rate.

real time, and when parameters are altered, it will interact with the patient to assert for consciousness. This has an advantage compared to a gaming system in that it could interact with the patient when detecting abnormal physiological parameters during the night, whereas the gaming device's functionality is limited to when the patient is awake. An integral system would hypothetically allow detection of all seizure types, including those considered subtle like absence seizures.

3. Types of seizure detection sensors

3.1. Unimodal detection devices

3.1.1. Electroencephalogram

Video-EEG has long been considered the gold standard for the diagnosis of seizures. Several groups have developed algorithms for automatic seizure detection based on EEG [26–31]. Most of them tested their algorithms with data from the Freiburg Seizure Prediction and the European Epilepsy Databases [26,27,30–34]. Most researchers used data from 2 to 6 electrodes trying to recreate the conditions for ambulatory performance. The Freiburg database uses three electrodes close to the focal area and three remote from it [26,27,32,34].

Researchers first differentiated the preictal period from the ictal period – which can be challenging as the transition is sometimes subtle [33,35]. The preictal period is seizure and patient specific, and its identification requires a training period of the algorithm for the individual patient [27,29,30,33,35]. Investigators developed an algorithm based on the parameters that yielded the best performance [29,30,32,35].

Seizures detected with this approach were focal dyscognitive seizures, focal without dyscognitive changes, secondarily generalized seizures, and absence seizures [27,28,33–37]. Intracranial EEG yielded a sensitivity of 80.5–98.8% and a false detection rate (FDR) of 0.054–1/h, within a prediction horizon of 30–60 min [26–31,34,35,38]. The data from scalp EEG had a sensitivity of 74–99% and an FDR of 0.28–1/h, with an anticipation time of 16 min [28,32,36,39,40]. A device for detection of absence seizures achieved a sensitivity of 95%, with an FDR of 0.037/h, when tested in an ambulatory setting in one patient [41]. The electrodes were placed in F7-Fp1, as this yielded the best results in a previous study [37].

Patients must wear scalp electrodes and remain attached to EEG equipment during monitoring, which increases artifact, is impractical, and potentially leads to stigmatization and discomfort [42]. The current devices for ambulatory EEG could be used as a model, as they are light weight and patients can wear them attached to a belt or over the shoulder [43]. Several researchers are working on the development of wireless EEG modalities, with few and small electrodes [44–49]. A group used four electrodes installed in a 20 cm area and demonstrated that this is feasible and comparable to a 10–20 EEG system [44]. The Emotiv EPOC is a low-cost wireless system with 14 electrodes and a battery life of 12 h enabling the device to be worn during the day [50,51]. The B-Alert is a wireless EEG head unit with 21 channels (20 electrodes), in which the data is transmitted to the receiver [50]. These two systems were compared and B-Alert (in two different versions) was the most similar to a research-grade system and it detected all of the EEG expected features [50,51]. Another wireless system was developed for ambulances and emergency departments with a short preparation time because it is a cap with six channels [52]. There is also a portable, wireless system that uploads the data to a smartphone [47]. It has two electrodes, one over the area of interest and the other over the mastoid, and it was able to record a seizure [47]. Finally a waterproof, 2 electrodes (1-channel) EEG monitoring device called the EEG Patch TM is able to track seizures for 7 days and allows home monitoring [53]. The experience with

these devices detecting seizures and their sensitivity compared to standard EEG is not yet published for some of them. They represent promising systems that when linked with an automated detection algorithm could allow for accurate seizure detection.

Alternatively, intracranial EEG is an invasive procedure that requires close supervision and is associated with risk of infection. In the future, devices with only a few and small electrodes will increase portability, and machine learning algorithms will further automatize the seizure identification process. Large studies targeting homogeneous seizure types with machine learning algorithms will clarify the role of EEG as a potential portable seizure detection device.

3.1.2. Surface electromyography (sEMG)

Most seizures have a motor component; therefore the analysis of muscle activity with sEMG is a viable option for seizure detection [54]. sEMG helps differentiate epileptic seizures from non-epileptic seizures: epileptic seizures have a large proportion of EMG activity in the frequency band above 100–150 Hz [55,56]. sEMG detects muscle activity with as few as one channel; deltoid and anterior tibialis muscles are the preferred placement sites [55–57].

Tonic stiffening consists of an intense muscle contraction, which allows for early GTCS detection [54,55,57]. In a study, the efficacy of deltoid muscle detection site was better than for the anterior tibialis muscle, with a sensitivity of 100%, mean detection latency (time from beginning of a seizure to its detection) of 13.7 s, and FDR of one alarm per 24 h [55]. When using only the nocturnal data, the FDR improved to one every 10 nights [55]. sEMG in the anterior tibialis muscle had a sensitivity of 57%, but the FDR improved to one false alarm in 12 days with a latency of 25 s [57]. The sEMG on tonic seizures, recorded at the deltoid muscle, had a sensitivity of 53–63% and a FDR of 1.49–4.03 [54]. A recent study placed the sEMG electrodes in the biceps and triceps detecting 95% of GTCS but none of the other seizures (myoclonic, tonic, absence, and focal seizures with or without loss of consciousness) [58].

Disadvantages of sEMG sensors include discomfort when strongly fixed to the skin and potential for detachment [57]. Better results may be achieved when certain parameters are tailored to the individual patient, especially for tonic seizures [54]. In summary, sEMG reliably detects GTCS and tonic seizures and can potentially detect other seizure types with a motor component.

3.1.3. Electrodermal activity (EDA)

Modulation in skin conductance is referred to as EDA, and it reflects the activity of the sympathetic branch of the autonomic nervous system [59,60]. Sweat increases the conductance of an applied current [59]. The device applies direct current to the stratum corneum beneath the electrodes, and measures the EDA in the ventral side of distal forearms [59,60]. The rationale for the use of this device is that epileptic seizures transiently increase EDA [60].

In a study with seven patients the EDA was significantly elevated immediately after the onset of each EEG seizure, including GTCS and focal dyscognitive seizures [60]. But the change in EDA was higher and remained elevated for a longer period in GTCS compared to focal dyscognitive seizures [60]. In a study including 11 patients, 100% of GTCS had a greater than 2 standard deviation (SD) increase in EDA, but only 86% of focal dyscognitive seizures had it, with a median latency for all seizures of 33 s [61].

Studies in adult and pediatric patients have demonstrated a strong correlation between the duration of postictal generalized EEG suppression (PGES) and the degree of EDA response. Children seem to have lower PGES duration, which could explain lower sudden unexplained death in epilepsy (SUDEP) rates, although

they have more sympathetic activation and diminished vagal tone when compared to adults [62,63]. EDA recordings may also be able to help us better understand the pathophysiology of SUDEP [61,62].

Disadvantages include that EDA recording is susceptible to motion and pressure artifacts and it could be uncomfortable or obtrusive. The EDA in the ventral side of the distal forearm is well tolerated, even for long periods [59]. Large studies on continuous ambulatory autonomic monitoring will provide insights to optimize this promising modality.

3.1.4. Electrocardiogram (EKG)

Cardiovascular changes are relatively easy to measure, and in patients with epilepsy they are particularly relevant as they may be linked to SUDEP [64,65]. EKG can be recorded from a single channel and has a higher signal to noise ratio than EEG [66]. Multiple studies have aimed to characterize heart rate changes before, during, and after seizures [64–70]. Heart rate changes could be explained by increased motor activity, release of catecholamines, sympathetic and parasympathetic shifts, activation of limbic structures, increased neuronal firing, or a combination of these and other unknown factors [64,66,69,71]. Another aspect that must be taken into consideration is that heart rate also depends on the state of vigilance so sensors using heart rate to detect seizures might be affected by this [72,73]. The pattern of heart rate changes seems to be patient-specific, reflecting the individual spread and evolution of seizure activity and warrant the development of patient-tailored detection algorithms [64,69,71,74].

EKG has been used to detect focal seizures, secondarily generalized seizures and GTCS [75]. EKG abnormalities have been linearly correlated to electrocorticogram (ECoG) seizure severity, proving the feasibility of EKG as a seizure detection device [67]. Ictal tachycardia was more prominent when arising from the right hemisphere [65,74]. In contrast, short myoclonic seizures often did not produce heart changes [64]. In a large study, 73% of focal seizures had heart rate increase, and in 23% it preceded EEG onset [68]. In another study in children, ictal tachycardia was present in 70% of focal seizures (temporal or frontal onset), but not in generalized seizures [75]. One interesting study found that heart rate increased from the preictal to the ictal period in 74% of the patients and it was higher in seizures with secondary generalization than in complex partial (focal dyscognitive) seizures [76]. One of the few studies that included generalized seizures detected heart rate changes in 35% of seizures, either GTCS or secondarily generalized seizures [65]. The sensitivity of the automated detection algorithms was 90–98% in seizures with heart rate changes, with a greater than 50% positive predictive value [64,77]. The latency was between 0.8 s and 10 min for all seizure types [66,69,71,74,77].

Regarding ambulatory monitoring, a group developed a wireless device for real time detection of seizures that allows unobtrusive monitoring with good mobility. This may be effective for detection of tonic-clonic, tonic, clonic and hypermotor seizures [78]. The monitor is composed of a wireless sensor board, ultra-low power EKG sensor readout, accelerometer, and micro-secure digital-card. The detection algorithm has a sensitivity of 99.8% and positive predictive value of 99.8% [79]. The device with two electrodes was placed on the left arm of four healthy volunteers who considered it comfortable and reported no inconvenience during sleep [78]. A smartphone application monitors heart rate remotely based on skin color changes [80]. Since cardiac pulse leads to subtle changes in skin color, a photoplethysmographic signal can be measured recording the face with the front facing-camera of a smartphone. Heart rates were measured for a minute with good accuracy in adults who held the smartphone 30 centimeters away from their face without moving [80]. This technique

could ultimately prove useful in patients with seizures having a motor component. Another option utilizes cardiac-based activation vagus nerve stimulation as part of a commercially available closed-loop system [81]. This cardiac-based seizure detection algorithm had a sensitivity of 80%. The effect on seizure frequency was moderate but there was significant improvement in QOL [81].

Some disadvantages of EKG as a seizure detection signal include the low specificity of changes in heart rate [64,82], low stability of electrodes, and discomfort with long-term use, but these could be avoided with wireless, video, or VNS activated devices. The findings above and the development of an ambulatory device demonstrate that automated EKG seizure detection is possible, particularly when parameters are tailored to the individual patient.

3.1.5. Accelerometry (ACM)

ACM has been used for motor seizures as it detects changes in velocity and direction [19,83–85]. The signal is recorded by means of a three-axis motion/accelerometer sensor, a microprocessor, and a small, rechargeable battery, usually placed on a limb [83,84,86–88]. The main challenge is to differentiate seizures from normal, daily, repetitive movements [19,89]. Some systems have a cancel button and this gives the opportunity to indicate that a movement was a false alarm, avoiding a false-positive alert to the caregiver [84].

This modality was able to detect focal seizures with minimal motor component, GTCS, secondarily generalized seizures, myoclonic, clonic, tonic and hypermotor seizures [19,23,83–85,87,90,91]. Clonic seizures present with a burst-like pattern, which was better identified and discriminated from other movements [19,90]. Tonic seizures are block-shaped because the acceleration is almost constant. They resemble slow normal movements, which makes them harder to identify [19,85]. Focal dyscognitive seizures without motor phenomena and absence seizures were not detected [19,83,84]. Sensitivity ranges between 16 and 100%; one study had a FDR of 0.2/day [83,84,86,87,89–92]. Seizures were detected 9–60 s after seizure onset [84,87]. The same accuracy for nocturnal and daytime seizures was achieved [19,83,84]. One study validated the system in a home environment detecting 78.5% of the seizures reported by parents, with 0.6 false alarms per night [92].

Some disadvantages include that the system is restricted to seizures with a motor component, and that seizures are not detected when there is an obstacle to free limb movement [83]. More studies in ambulatory settings are needed, as most studies have been in epilepsy monitoring units where movements might be limited [83,87]. This modality has good sensitivity with good night detection rates, and most patients and families found the device user-friendly [83,84,90].

3.1.6. Video detection systems

Automatic video detection systems use velocity, area, duration, rotation, oscillation, angular speed, and/or displacement (motion trajectory) to detect seizures [93–97]. The underlying concept is to detect complex motor patterns by automatic interpretation of video data [93]. The systems have been classified as marker-based or marker-free, depending on whether the cameras track detectable markers placed in relevant places [93].

Seizure types that can be captured by video include focal, hypermotor, myoclonic, and clonic [93–95,98]. Myoclonic seizures are detected with good sensitivity and specificity with a marker-based system using spatio-temporal interest points [95]. Reference markers could be placed on the head, trunk and extremities to assess for movement when evaluated with infrared light by a video system. This was done with frontal and temporal lobe seizures to evaluate for the lateralizing value of ictal head turning [99,100]. The overall sensitivity varies from 75 to 100%, positive

predictive value over 85%, and specificity of 53–93% [93–96]. Reported latencies range from 4.6 to 21.4 s [94].

Marker-based devices present with the shortcoming that sensors can be uncomfortable or dislocate over time [94,97]. Marker-free systems detect only seizures with a motor component, and they are more limited to the area covered by video: the patient must be visible and properly placed [93]. Seizure detection based on video is feasible, but it recognizes mainly seizures with large movements.

3.1.7. Mattress sensors

Mattress sensors were developed because most cases of SUDEP occur in patients with GTCS while unsupervised in bed at night [101–105]. These devices consist of a sensor placed under the patient's mattress and connected to a monitor. The sensor alerts the family when it detects a stimulus above the set threshold [102,103,105]. The MP5 bed seizure monitor detects movement and noise frequency and intensity, with a minimum patient weight of 25 kg [102]. The ST-2 bed alarm mainly detects abnormal bed motion in patients weighting more than 6.4 kg [103]. The Emfit monitor detects micro-movements, without a minimum weight [105].

For GTCS during sleep the MP5 sensitivity was 62.5% and specificity 90% [102]. In another study including multiple seizure types, the device only detected one GTCS during sleep (16.7%) [103]. The ST-2 did not detect any of 20 nocturnal seizures. It only detected one focal dyscognitive seizure with motor phenomena of the 26 seizures recorded in awake patients, with an overall sensitivity of 2.2%. These two devices were designed to detect GTCS, but their sensitivity is suboptimal for seizures overall [103]. The Emfit sensitivity was 30% during daytime and 85% during sleep for GTCS [105].

Most of these devices have the disadvantages of having a weight restriction, detecting only seizures with rhythmic movements, and a low sensitivity [103,105]. Mattress devices allow for adjustment of the parameters to compensate for individual differences in movement during sleep [103]. Individual calibration and testing over a couple of nights in a home setting are recommended [102]. None of these systems performed as well as video EEG in detecting nocturnal seizures but this is a step in the right direction for a much needed device.

3.1.8. Seizure-alert dogs

Seizure-alert dogs are those that display some attention-getting behavior prior to human appreciation of an impending seizure event [106–108]. While the trigger to which these animals respond is not completely understood, it is believed that they alert to subtle human behavioral changes [106,107,109]. Hypothetically they may be responding to changes in human respiratory or heart rate or even olfactory phenomena, perhaps pheromone production [106,108]. They are able to alert from 30 s to 45 min before seizure onset [108–110].

Seizure-alert dogs have been reported to detect atonic, focal dyscognitive, and GTC seizures [106,110,111]. One of the few seizures detected by a seizure-alert dog with concomitant video EEG monitoring was focal dyscognitive [111]. One study reported a median sensitivity estimate of 80% and specificity of 100% [107]. Another study found a 43% mean reduction in seizure frequency with the use of seizure-alert dogs, hypothetically due to diminished stress [109].

There are few reports of EEG or other monitoring concomitant to the dogs' behavior and most articles used diaries or questionnaires as the gold-standard, raising some skepticism on the reliability of the information [111,112]. Other difficulties are that seizure-alert dogs also respond to psychogenic seizures, as reported on at least seven well documented cases, and they are

not able to monitor patients during their own sleep [82,111,113]. Companionship seems to be one of the primary benefits and this could lead to the increase in QOL [110–112]. To date, no rigorous data confirms whether seizure prediction by seizure-alert dogs is better than chance [106].

3.1.9. Cerebral oxygen saturation sensors

Increased cerebral blood flow precedes the onset of a clinical seizure in temporal lobe epilepsy by approximately 20 min [114]. This finding was confirmed by single-proton emission computed tomography (SPECT) [115]. One study placed transcutaneous regional cerebral oxygen saturation (rSO₂) sensors on each side of the forehead in five patients with GTCS, and compared this to video-EEG. The mean rSO₂ value significantly increased in the preictal period by at least 3 SDs in 4 out of the 7 registered GTCS, on average 18 min prior to EEG seizure onset [116].

Larger studies will clarify the relevance of this approach in other seizure types. These findings suggest that cerebral oxygen saturation sensors could have a role as automatic prediction devices, at least in GTCS, with a prolonged time of detection before seizure onset [116].

3.1.10. Near-infrared spectroscopy (NIRS)

Wavelengths used in NIRS measure cerebral oxygen saturation by using the specific absorption properties of tissues in the near infrared range [117,118]. The spectrophotometer, typically placed on the forehead [117–119], emits light into the tissue from the surface of the scalp, and then collects it from a detector close to the emitter [118].

Seizures analyzed with this approach are mainly focal, including focal dyscognitive, focal without impairment of consciousness, and focal with secondary generalization, but also absence seizures [117,119]. One study in temporal lobe seizures found an increase in rSO₂ in the preictal period, followed by a decline around seizure onset, and then a postictal increase [117]. The mean preictal rSO₂ increase was 7% above baseline, and 5 min before seizure onset [117]. A study in children with generalized spike and wave discharges had similar results [118]. The authors revealed that frontal generalized spike and wave discharges were associated with moderate oxygenation 10 s before the discharge, followed by a strong deoxygenation, then a second increase in oxygenation, and a return to baseline [118]. In another study, the hemodynamic changes in the frontal lobe during seizures were only slightly higher than during non-seizure activities [119].

One of the disadvantages of this system is that the sensor pad might be difficult to wear from a cosmetic point of view, since it must be surrounded by a black cloth so that no external light interferes with the NIRS signal [119]. In addition, one of its biggest challenges is developing a generic algorithm, since there is a huge diversity in hemodynamic changes amongst patients [119]. The findings suggest that NIRS might be a noninvasive detection system, at least for focal seizures, with a patient specific algorithm [117].

3.1.11. Implanted advisory system

This system was developed to predict and quantify seizures in adults with refractory focal seizures [120]. Two silicon leads, each with eight contacts, were placed over the quadrant suspected to contain the epileptogenic zone [120]. A hand-held device analyzed the recorded EEG based on a patient specific algorithm, developed during the data collection phase. This device transmitted audible and visual signals showing the likelihood of seizure occurrence minutes to hours before it actually occurred [120].

The mean sensitivity after four months of implantation was 66% (10 patients). The usefulness of seizure prediction was inconclusive, and the variability in warning times and difficulties adapting to the system seemed to play a role [120].

The main difficulty with this system is that it requires an invasive procedure that could have complications [120]. In one year after implantation, four patients (27%) had serious adverse effects, including device migration and infection. This study demonstrated for the first time that a seizure detection device predicted seizures better than chance [120].

3.1.12. Skin temperature

The ratio of mean temperature during sleep and awake periods could be related to seizures over the following days as suggested by recent work with wristband data in children with epilepsy. This work is in progress but if confirmed it suggests that skin temperature could also be analyzed for seizure detection [121].

3.1.13. Respiratory monitor

Cardiac and respiratory abnormalities have been suggested as a cause for SUDEP [122–124]. In a study with 26 children (101 seizures) an elastic belt was used to monitor chest and abdominal excursion by respiratory inductance plethysmography [122]. Of the seizures 39% were associated with ictal central apnea, 34% with ictal tachypnea, and 13% with ictal bradypnea. No patients had ictal obstructive sleep apnea [122]. Ictal apnea was more frequent in temporal lobe seizures when compared to frontal seizures (OR 8.04, $p = 0.0005$). A thoracic band for detecting cardiac

and respiratory changes could be coupled with a movement detection modality, potentially improving the overall sensitivity.

3.2. Multimodal detection devices

Seizure detection is more accurate if it combines more than one modality, as multimodal systems have shown increased sensitivity and lower FDR [125–128].

3.2.1. EDA and ACM

The combination of EDA and ACM might improve detection of motor seizures and those with autonomic involvement. A system with EDA and ACM biosensors-placed on the ventral forearms of six patients with GTCS, during awake and sleep states – yielded 94% sensitivity and one false alarm per 24 h. The FDR was lower than for ACM alone [129]. The mean latency from clinical onset to detection was 31 s [129]. This emerging modality may benefit from more training in different settings, and preliminary results are promising.

3.2.2. sEMG and ACM

This combination yielded better results than each signal individually, and it also enhances detection of seizures with motor and autonomic system involvement. The modality was proposed

Table 3
Comparison of the available seizure prediction sensors.

Seizure detector	Primary input	Seizures detected	Advantages	Disadvantages
EEG	EPSP/IPSPs	Focal dyscognitive seizures, focal without dyscognitive changes and secondarily generalized, absence seizures	Gold-standard. Good sensitivities were achieved	Discomfort, stigmatization due to electrodes. Risk for movement artifact
Intracranial EEG	EPSP/IPSPs	Focal dyscognitive seizures, focal without dyscognitive changes and secondarily generalized	Gold-standard. Good sensitivities, were achieved	Invasive procedure with potentially life-threatening complications
sEMG	Muscle movement	GTCS, tonic seizures, hypermotor seizures	Early detection in tonic phase with one channel. Good sensitivity and nighttime detection	Only detects seizures with a motor component. Electrodes could detach, move or be uncomfortable
EDA	Sweating	GTCS, Focal dyscognitive seizures	Good detection rates, including FDS, better when combined with ACM	Susceptible to motion and pressure artifacts, could be uncomfortable
EKG	Heart rate changes	Focal seizures, secondarily generalized seizures and GTCS	Heart rate is relatively easy to detect. Can be recorded from one channel. Higher signal to noise ration than EEG	Seizures without HR changes go undetected. HR changes occur in many everyday activities and seem to be patient specific. Electrodes might be uncomfortable or unstable on the long-term.
ACM	Movement	GTCS, secondarily generalized, myoclonic, clonic, tonic and hypermotor seizures	Great sensibility, good night detection rates. User friendly.	Only detects seizures with a motor component and when there is free limb movement. Battery system needs to be optimized
Video	Movement	Focal, hypermotor, myoclonic and clonic	No discomfort for patients in marker-free video	Mainly detects seizures with a motor component. Limited to the area covered by video, the patient must be visible and properly placed. Attached markers could dislocate or produce discomfort
Mattress sensor	Movement, noise	GTCS, Focal dyscognitive seizures with motor phenomenon	Allows nocturnal GTCS detection. There is no need to place electrodes on the patient. Parameters can be adjusted for every patient	Only detects seizures with rhythmic movements. There is a weight limitation. Some find it uncomfortable. Low sensitivity.
SADS	Subtle behavioral changes are detected by the dogs	Focal dyscognitive seizures, GTCS	Alert before the seizure. Increase in QOL, possible reduction in seizure frequency	Few studies while the patient is on EEG. They also alarm to psychogenic seizures. Cannot detect while dogs sleep
Cerebral oxygen saturation sensor	Cerebral blood flow	GTCS, temporal lobe seizures	Can detect seizures up to 18 min before clinical onset	Few studies, only in 2 seizure types
NIRS	Regional cerebral oxygenation	Focal dyscognitive seizures, focal without dyscognitive changes and focal with secondary generalization, absences	Good detection, including FDS	Big diversity in hemodynamic changes. Electrode could be uncomfortable and must be surrounded by a black cloth
Implanted advisory system	EPSPs/IPSPs	Focal seizures	Gold-standard. Good seizure prediction in an ambulatory setting	Invasive procedure with potentially serious adverse effects

EPSP: excitatory post-synaptic potentials. IPSPs: inhibitory post-synaptic potentials.

because ACM seems to be more sensitive in detecting the clonic phase, and sEMG the tonic phase of seizures [126]. In a study of children with tonic-clonic seizures, the combination of two ACM and two sEMG sensors achieved a sensitivity of 91%, with a FDR of 0.5/12 h and a latency of 10.5 s [126]. The FDR was higher than previous studies on adults, which probably reflects the fact that children usually move more than adults at baseline [126]. The ACM sensors yielded best results on the left wrist (non-dominant hand) and the right ankle [126]. The use of isolated sEMG yielded a slightly lower sensitivity (82% vs 86%) when compared to isolated ACM, but FDR and latency were lower [126]. sEMG alone detected tonic and hypermotor seizures, while ACM detected hypermotor, tonic, and clonic seizures [126].

3.2.3. Magnetometer and ACM

The magnetometer is used as a seizure detection sensor because it characterizes movement in the horizontal plane, as seen in a tonic seizure [130]. Magnetometers measure body inclination using a 3D Earth's magnetic field sensor and track changes in orientation in 3D space [131]. Three sensor modules, each containing a tri-axial ACM and a tri-axial magnetometer, were placed in the head and both wrists of patients [130]. In a first phase, sensors discriminated tonic activity, clonic activity, hypermotor movements, and no movements [130]. Afterwards the system was tried in 86 patients in an epilepsy monitoring unit [132]. There was accurate classification of tonic seizures in 62% of cases and tonic-clonic in 90% of the cases [132]. Hypermotor seizures were classified as tonic-clonic seizures in 100%. The detectors had a 80% sensitivity with 95% specificity [132].

3.2.4. Video, ACM and radar-induced activity recording (VARIA)

Radars use wave transmission to identify motion caused, for example, by a change in the patient's position in bed. The combination of radar, video and ACM may increase the ability to detect seizures. This multimodal system considers the top 10% of events per night with the most abnormal movements, regardless of the number of seizures that occurred [23]. VARIA was tested in an 8-year-old patient with focal epilepsy with secondarily generalization. This system achieved a sensitivity of 56% and 20 false alarms per night [23]. The system detected some seizures that were missed by the staff [23]. Limitations of this system include numerous, easily removable bracelets [23].

3.2.5. Multi-modal intelligent seizure acquisition (MISA) system

The MISA system includes sEMG, magnetometers, ACM, and gyroscopes allowing full body movement description. Gyroscopes provide information on the rotation of each joint [125]. The system consists of a suit with 16 sensors, each one containing 3D accelerometer, 3D magnetometer, and 3D gyroscope, as well as 28 sEMG electrodes placed on 14 muscles [127]. The MISA system was first tried on three subjects that simulated tonic-clonic, versive asymmetric tonic, and myoclonic seizures [127]. The system achieved the best results when all modalities were used [127]. The non-specific system detected 98% of simulated seizures with one false positive per hour, and the subject specific system detected all seizures, with only one false positive in 4 h [127]. On another study the MISA system had the best results followed by the combination ACM and sEMG [125]. Some patients found the suit uncomfortable so the prototype is under modification, probably involving smaller and fewer electrodes [125,128].

3.2.6. EEG and EKG

A group developed a computerized system that releases a vagus nerve stimulator (VNS) discharge when a pre-specified event (seizure or epileptiform discharge) is detected [133]. This system collects information from EEG and EKG and then a computer

analyses it in real time, using patient specific algorithms [133]. In a previous study this EEG automated algorithm detected 92% of seizures, with a 6.8 ± 2.4 s detection latency and 0.2 ± 0.7 false alarms per hour [134]. An adapted algorithm detects acceleration in heart rate associated with seizure onset [133]. When the system detects the onset of a pre-specified event it energizes the electromagnet worn by the patient. This algorithm detected focal dyscognitive seizures, secondarily generalized seizures (in the tonic-clonic phase), and epileptiform discharges [133]. This study demonstrates that closed-loop systems are feasible and potentially effective.

4. Correlation between sensor and seizure type in the literature

There have been numerous studies aiming to find an adequate seizure detection device, but so far few have been guided by patient characteristics and seizure types. Abundant evidence confirms that patient specific algorithms are crucial for achieving accurate detection devices [54,64,69,71,74,94,95,102,119]. Different sensors have been used for detecting GTCS, including mattress sensors, EDA, sEMG, ACM, cerebral oxygen saturation sensors and seizure-alert dogs with promising results [55,57,58,60,61,65,83,84,90,102,103,105,132]. However, there is limited literature for seizure detection rates in other seizure types. Tonic seizures have been detected with sEMG and ACM [54,64,87]. For clonic seizures there are some studies with ACM and video detection [96]. For hypermotor seizures ACM, sEMG, and video have demonstrated good detection rates [38,86,98]. For myoclonic seizures ACM and video had good detection rates [85,95]. Focal dyscognitive seizures

Table 4
Sensitivity for each sensor (overall and by seizure type).

Seizure detector	Overall sensitivity	Sensitivity by seizure type
Intracranial EEG	Intracranial 80.5–98.8%.	Focal seizures 80–98.8% Mesial temporal 80.5%
Scalp EEG	Scalp: 74–99.1%	Focal seizures 74–96.6% Absences 95–99.1% (95% ambulatory setting)
sEMG	53–100%	Tonic seizures 53–63% GTCS (deltoid muscle) 100% GTCS (tibialis muscle) 57% GTCS (biceps/triceps) 95% GTCS 100%
EDA	86–100%	Focal dyscognitive seizures 86%
EKG	35–99.8%	Tonic + myoclonic seizures 48% Focal seizures 70–99.8% Generalized seizures (GTC + secondarily generalized) 35%
Accelerometry	80–100%	GTCS 87.5–100% Hypermotor seizures 95.71% Myoclonic seizures 0–80% Focal with minimal motor component 2% Tonic 10% Motor/Hypermotor seizures 93.3–100% Myoclonic seizures 75% Focal dyscognitive seizures 57%
Video detection systems	75–100%	
Cerebral oxygen saturation sensors	57%	
Mattress sensor	0–75%	GTCS 0–75% Focal dyscognitive seizures 0–16.6% Tonic + myoclonic 0–8.3% Tonic 27% Myoclonic 0% GTCS 80%
Seizure-alert dogs	80%	Focal seizures 66%
Implanted advisory system	66%	

NIRS (near infrared spectroscopy): data not available.

Legend: ACM: accelerometry. EDA: electrodermal activity. EEG: electroencephalogram. EKG: Electrocardiogram. GTCS: Generalized tonic-clonic seizure. sEMG: surface electromyogram.

Table 5
Currently available devices.

Company	Brand name	Device type	Website
ActiGraph	wGT3X, wActiSleep, GT3X, ActiSleep	Triaxis accelerometer watch	http://www.actigraphcorp.com
Advanced Brain Monitoring	X series – EEG wireless monitoring B-Alert wireless EEG systems	EEG wireless headset for interpretation of physiological signals	http://advancedbrainmonitoring.com
Affectiva	Affectiva Q-Sensor	Wearable, wireless biosensor that measures EDA, temperature and activity	www.affectiva.com
Air Brain System/Kwansei Gakuin University	Air Brain System	Portable EEG telemetry system using 3G network with a smartphone	http://eudl.eu/doi/10.4108/icst.bodynets.2013.253918
Alert-It	Ep-It Companion Monitor (S1029) Ep-It Guardian Monitor (P139) Falls Detector	Bed motion monitor, accelerometer located under mattress with panic button	http://www.alert-it.co.uk/epilepsy/
Apple Inc.	EpiWatch	Apple Watch App works with Health App to measure heart rate, accelerometer and gyroscope data collected by iPhone and Apple Watch	https://itunes.apple.com/us/app/epiwatch/id1047757228?mt=8
Ashametrics Company	Wrist LifeBand, Ankle LifeBand, Chest LifeBand	Soft, wearable textile bands that measure EDA, accelerometer, and temperature. They transmit data wirelessly to a mobile phone or nearby PC and records data internally that can be downloaded later.	http://www.ashametrics.com/products
Basis	Peak Fitness and Sleep Tracker	Measures heart rate, rapid eye movements sleep, movement, skin temperature and galvanic skin response	http://www.mybasis.com/
BioLert	EpiLert	Watch-like sensor used as an epilepsy alert and treatment aid system for home care	http://www.biolertsys.com
Baby Ping	Baby Ping Video/Audio Monitor	Baby video and audio monitoring device including night-vision capabilities	www.babyping.com
The Bhutan Epilepsy Project/Grand Challenges Canada	2014 The Bhutan Epilepsy Project	Portable EEG telemetry system using 3G network with a smartphone	http://www.bhutanbrain.com
Capture Proof	Capture Proof	HIPAA compliant platform to share medical videos and pictures via online application	www.captureproof.com
Cyberonics Inc.	103 Demipulse [®] 104 Demipulse Duo [®] 105 AspireHC [®] 106 AspireSR [®]	Implanted VNS Therapy system The new version of the VNS detects seizures based on heart rate changes	http://us.livanova.cyberonics.com/en/vns-therapy-for-epilepsy/healthcare-professionals/vns-therapy/about-products/
Danish Care ApS	Epi-Care [®] free Epi-Care [®] 3000	Armband with acceleration sensors to detect convulsions and vibrations. Alarm for detecting epileptic seizures during sleep	http://danishcare.dk/uk
D.C.T.Associates Pty Ltd.	Vigil-Aide	Registers initiation of seizure through movement detection, triggers an alarm which is audible, vibratory, or visual	http://www.dctassociates.com.au/convul.htm
Electramed Corporation	Pro-Tech zRIP Kit	Respiratory Inductive Plethysmography system including sensor belts and driver module Provides true breathing movement tracing	https://www.electramed.com/neurology/sleep-sensors/respiratory-belts/nr37001808-pro-tech-zrip-kit-adult
Emfit	Emfit Seizure Monitor	Bed motion sensor with accelerometer under mattress	http://www.emfit.com
Empatica	E3 Wristband E4 Wristband Embrace	Wristband using accelerometer, EDA, gyroscope, and temperature sensors to detect seizures Free mobile phone application Embrace is the only waterproof version	https://www.empatica.com
EpiCall Ltd.	EpiCall	Sticker placed on the side of the face with electrooculography and photoplethysmograph electrodes	http://clinicaltrials.gov/ct2/show/NCT01436695
EpDetect	EpDetect	Accelerometer based free mobile phone application to detect epileptic seizures with automated carer alert	http://www.epdetect.com
FitBit	Flex Charge Charge HR Surge	GPS capable, heart rate, sleep, and activity monitoring wristband	https://www.fitbit.com/surge
Garmin	Garmin Forerunner 310X	GPS capable watch	http://www.heartratemonitors.com

Table 5 (Continued)

Company	Brand name	Device type	Website
Hexoskin	Smart Shirt	Measures heart rate, respiratory rate, activity, sleep, and breathing volume	http://www.hexoskin.com/pages/key-metrics-delivered-by-hexoskin
Holst Centre/IMEC Hobo Heeze BV	Epilepsy Seizure Monitoring System Wearable sensors	Prototype that detects epileptic seizures. Integrated algorithm with wearable ultra-low power EEG, EKG and accelerometer sensors	http://www.hoboheeze.nl/engels/episode.html
IctalCare A/S	IctalCare 365	sEMG in the form of a body-worn “ePatch” attached to the upper arm	http://ictalcare.com/
JS Mobile	FaceBEAT	iPhone, iPad, and iTouch application that measures heart rate through the camera	https://itunes.apple.com/us/app/facebeat/id551645909?mt=8
Medpage	MP5, MP2, ST2	Bed motion sensor and vocalization microphone (accelerometer under mattress and microphone). Bed motion sensor (accelerometer under mattress)	http://www.medpageusa.com
Mio Alpha	Mip Alpha Strapless	Heart rate monitor	http://www.alphaheartrate.com
Misfit	Beddit Sleep Monitor Flash	Mattress strap to fit entire width, measures respiration, heart rate, movement and snoring	http://misfit.com/
Movisens	EDA and Activity Sensor edaMove	Wristband with electrodes that measure EDA and 3D acceleration	http://www.movisens.com
Neuro:On	Neuro:On Smart Sleep Mask	Measures brain waves (EEG), muscle tension (EMG), eye movement (EOG), pulse (pulse oximetry), body temp, and body movement during sleep (actigraphy)	https://neuroon.com/
NeuroPace	RNS [®] Neurostimulator	Small, battery powered device surgically implanted in the skull, delivers brief pulses of stimulation when it detects activity that could lead to a seizure	http://www.neuropace.com/product/overview.html
Nike	Fitness Tracking Sports T-Shirt	Smart shirt capable of tracking heart rate, blood pressure, hydration levels and skin temperature	http://www.globalsources.com/gsol/1/Smart-watch/a/9000000132786.htm http://qz.com/315924/the-patented-nike-shirt-that-could-track-your-heart-rate-and-blood-pressure-while-you-exercise/ http://news.rice.edu/2013/05/13/rice-has-help-for-people-with-epilepsy-2/
Rice University	SMART (Seizure Monitoring and Response Transducer) Belt	Belt detects increased electrical conductance in the skin and change in respiration rate	http://news.rice.edu/2013/05/13/rice-has-help-for-people-with-epilepsy-2/
SAMi	SAMi sleep activity monitor SAMi-3	Sleep activity monitor with video and audio to detect unusual motion disrupting sleep setting off an alerting alarm	http://www.samialert.com/sami-3/
Sensorium	Sensealert-102/EP200	Bed motion sensor (accelerometer under mattress)	http://www.sensorium.co.uk
Sparkfun	ADLX330	Small, thin, low power, complete three axis accelerometer wristband	https://www.sparkfun.com
Smart Monitor Corp.	SmartWatch	Motion detection and GPS capable wristband	http://www.smart-monitor.com
Polar	H1, H2, H7, FT1, FT2, Ft60, FT80, FT40, FT7	Heart rate, GPS capable, and waterproof watch	http://www.polar.com
Shilene	Seizure Alert and Recorder	Free mobile application for seizure detection (accelerometer currently under development)	http://shilene.com/
Suunto Timex	M5, Suunto Quest, Ambit3 Timex IRONMAN [®] Easy Trainer [™]	Heart rate and GPS capable watch Heart rate watch	www.suunto.com www.timex.com
Vahlkamp	Epi-Watcher	Bed motion sensor (accelerometer under mattress) with built in timer to avoid false positives. Contains a wireless alarm bell and a spoken message to pre-programmed numbers	http://www.vahlkamp.nl/Epi-Watcher_gb.html
Vicon	Vantage Motion Capture System Versions V16, V8, and V5	Camera system that detects motion, thermal changes, and accelerometry	http://vicon.com/products/camera-systems/vantage
Zephyr	BioHarness [™] 3 Chest Strap BioHarness [™] 3 Compression shirt	Measures heart rate, breathing rate, EKG, posture, and activity level	http://www.zephyranywhere.com/products/bioharness-3

Legend: EDA: electrodermal activity. EEG: electroencephalogram. EKG: Electrocardiogram. sEMG: surface electromyogram. GPS: global position system, VNS: vagus nerve stimulation.

were detected using EEG, implanted advisory systems, EKG, EDA, NIRS, seizure-alert dogs, and also by mattress sensors if they have a motor component [67,111]. Absence seizures were detected with EEG and NIRS [37,41]. This information is summarized in Table 3,

along with the advantages and disadvantages of each sensor. The sensitivity for each sensor, overall and by seizure type, is presented in Table 4. Also, there are several commercially available devices which are presented in Table 5. Overall, apart from GTCS, there are

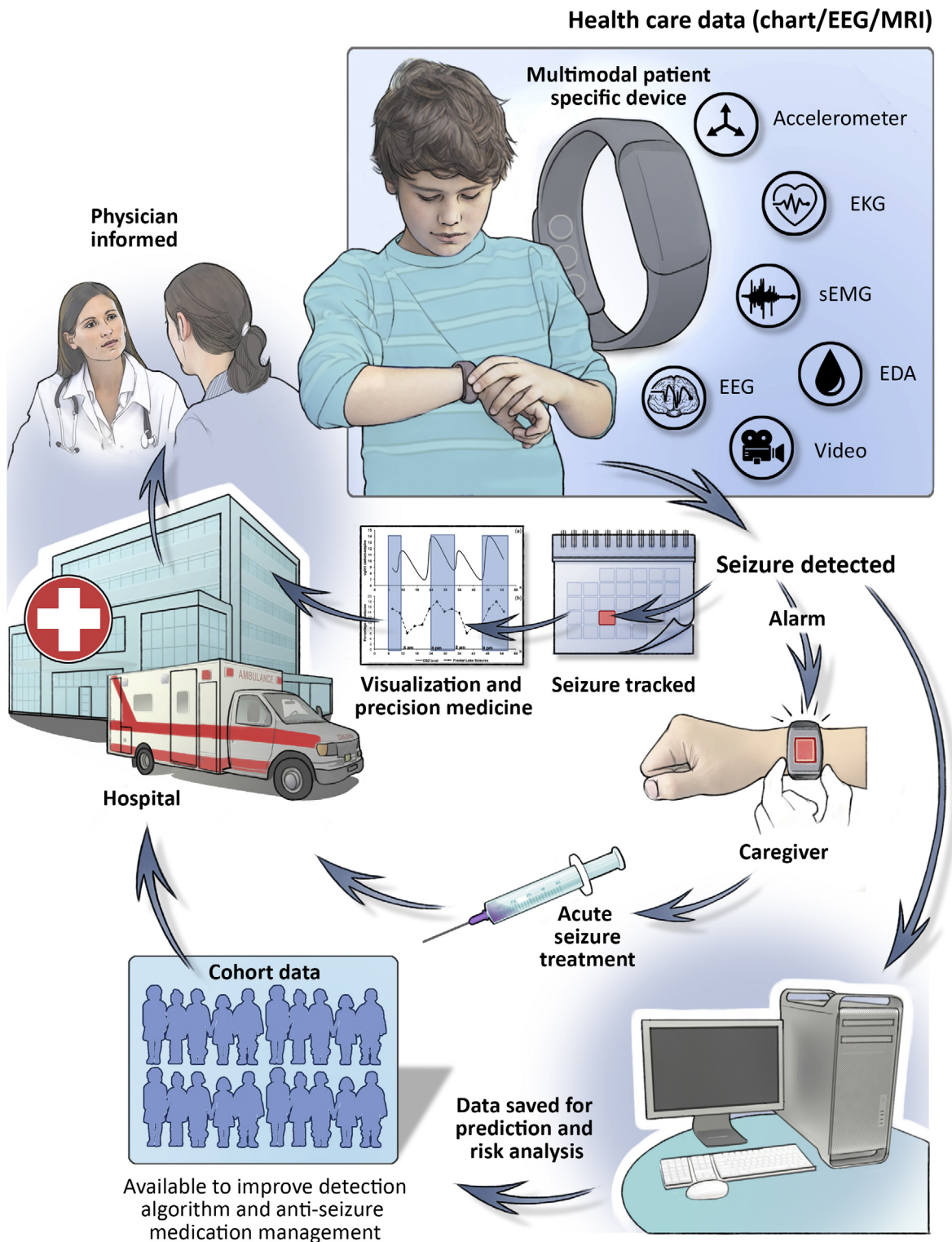


Fig. 1. Components of the seizure detection and treatment loop. The loop works with the patient's available health care data, including the clinical chart with the seizure types, the previous EEGs and the MRI results in order to select the best patient-specific combination of detection modalities. Some of the available detection modalities include accelerometer (ACM), electrocardiogram (EKG), surface electromyogram (sEMG), electroencephalogram (EEG), electrodermal activity (EDA) and video monitoring. Once the device is active data will be continuously collected from the patient and transmitted and saved for prediction and risk analysis. Then this information will be available to improve the detection algorithm and medication management based on large cohort information (external loop). All of these data will also be available to the health care system. When a seizure is detected the caregiver is informed and this leads to a corrective response which could imply abortive pharmacotherapy, neurostimulation or a micro-pump that delivers medication and transport to an emergency room, if needed (middle loop). Also when a seizure is detected it is tracked and available for visualization and precision medicine, analysis of previous treatment response and seizure prediction (internal loop).

few specific sensor detection options for each specific seizure type, and this is an impending necessity for patients and families.

5. Future directions

There is limited information on which sensor detection devices are optimal for each seizure type. We address this gap in knowledge by providing a guide on how to select specific detection devices for individual seizure types.

The ideal monitoring device should be safe and easy to use for patients, families and medical staff. It must be comfortable for patients, especially during sleep, and thus preferably wireless, miniaturized, and light weighted. If electrodes are used, they must be as small and few as possible since a significant percentage of patients are not willing to wear electrodes on a long term basis [42]. The device should be unobtrusive and discrete. Uncomfortable cables, electrodes, lights, buttons, and sounds should be avoided as this disturbs the patient and family, even more so in the long term. This was confirmed by a recent survey evaluating patients' desires that revealed strong preference for a seizure detection device that has little interference with daily activities [135].

Devices may either present with onboard processing and algorithms or may transmit non-processed signals. Onboard processing allows for longer battery life during real-time analysis and an alarm to be triggered when an event is outside of the chosen parameters. The alarm alerts caregivers of an impending seizure, permitting proper management and treatment and allowing for the development of closed loop systems. A cancel button for this alarm seems optimal so the patient can annul false alarms caused by everyday activities [84]. The cancel button should be appropriately disabled when the patient is not cognitively able to understand its mechanism. The system ideally transmits wirelessly and sends only concerning events to the control unit, in order to consume less energy, as done in some studies [83]. Low power consumption is a crucial issue, as some events could be missed if the battery's life span is too short [136]. Also, the device should contain a memory card that saves all information, and the caregivers could connect the device in a periodic manner so that the information is sent to the control unit.

These systems will improve and evolve as information from different patients accumulates. Cohorts of patients with similar characteristics will be informative for algorithm development applicable to increasingly specific patient profiles. The system should contain clinical information (past medical history, family history, medications, dose, side effects and other interventions), with specification as to these conditions when the EEG was obtained and how did they evolved in response to interventions. Such cohort information would allow for improvement in the seizure detection algorithm, as well as clinical management and medication selection based on increasingly large patient data sets, this is illustrated in Fig. 1. Ultimately, clinical care will improve, along with increased sensitivity and lower false detection rates of the devices and detection systems.

Closed-loop systems provide active feedback. They involve monitoring of patient specific parameters and respond to maintain these parameters within a range. After the data is acquired it must be transmitted for processing and analysis in a wireless and encrypted manner [82]. Closed-loop systems in epilepsy are a promising approach that may allow for rapid treatment in case of seizure detection [82,133]. This could be done with a stimulator that delivers a current (VNS, deep brain stimulator) or a micro-pump that delivers medication in order to abort the seizure [43,133,137–139]. In case of a seizure, the system could alert the caregiver or healthcare provider, further ensuring prompt intervention [82], as illustrated in Fig. 1.

Table 6

Proposal of seizure detection devices for each seizure type.^a

Seizure	First option	Second option	Third option	Fourth option
Tonic	sEMG	EDA	ACM	EKG
Clonic	ACM	sEMG	EDA	EEG
Myoclonic	ACM	sEMG	EDA	EEG
Atonic	EKG	EEG	Video	sEMG
Epileptic spasm	sEMG	ACM	EEG	Video
Hypermotor	ACM	sEMG	Video	EDA
Generalized Tonic-Clonic	sEMG	EDA	ACM	EKG
Focal dyscognitive seizure	EEG	EKG	EDA	NIRS
Autonomic	EDA	EKG	EEG	NIRS

^aSubjective, non-evidence based.

EEG refers to automated EEG detection

Legend: ACM: accelerometry. EDA: electrodermal activity. EEG: electroencephalogram. EKG: Electrocardiogram. GTCS: Generalized tonic-clonic seizure. HR: Heart rate. NIRS: near infrared spectroscopy. sEMG: surface electromyogram.

Regarding the selection of a specific device, patient characteristics and seizure types should be evaluated and individual parameters should be prioritized for the development of algorithms [134]. An initial training phase should be implemented to tailor the algorithm to the patient's characteristics, seizure related, and non-seizure related data. Table 6 contains a proposal of options for detection devices for each seizure type. The optimal device for each seizure type may include a combination of sensor modalities, since multimodal devices have yielded the best results.

6. Conclusions

Despite an internationally active research effort, there are limited data on which is the best sensor for each seizure type. This review provides a proposal in order to select the best sensor for each individual patient. It is likely that multimodal patient-specific detection systems will be needed to meet the complex requirements of seizure detection.

Author contributions

Adriana Ulate-Campos performed the literature search and literature revision, participated in study design, drafted the manuscript, reviewed, and edited for important intellectual content.

Francesca Coughlin participated in the interpretation of data in the literature, reviewed, and edited the manuscript for important intellectual content.

Marina Gaínza Lein participated in the interpretation of data in the literature, reviewed, and edited the manuscript for important intellectual content.

Iván Sánchez Fernández participated in literature search and literature revision, participated in study design, drafted the manuscript, reviewed and edited for important intellectual content.

Phillip L. Pearl participated in the interpretation of data in the literature, reviewed, and edited the manuscript for important intellectual content.

Tobias Loddenkemper developed the original concept and study design of the manuscript, participated in the interpretation and analysis of data in the literature, reviewed, and edited the manuscript for important intellectual content.

All authors gave approval to the final version of the manuscript to be submitted and all authors are in agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Conflict of interests

Adriana Ulate-Campos reports no disclosures.

Francesca Coughlin reports no disclosures.

Marina Gaínza Lein reports no disclosures.

Iván Sánchez Fernández is funded by a grant for the study of Epileptic Encephalopathies from “Fundación Alfonso Martín Escudero” and by the HHV6 Foundation.

Phillip L. Pearl is funded by NINDS (R01 NS 82286) for the study of 4-hydroxybutyric aciduria and by the SSADH Foundation, and receives royalties from Demos Medical Publishers, Up-to-Date, and Medlink. He serves as Associate Editor for the *Journal of Child Neurology* and on the editorial boards of the *Annals of Neurology*, *Neurology*, *Future Neurology*, and *Music and Medicine*.

Tobias Loddenkemper serves on the Laboratory Accreditation Board for Long Term (Epilepsy and Intensive Care Unit) Monitoring, on the Council (and as Treasurer) of the American Clinical Neurophysiology Society, on the American Board of Clinical Neurophysiology, as an Associate Editor for *Seizure*, as Contributing Editor for *Epilepsy Currents*, and as an Associate Editor for *Wyllie's Treatment of Epilepsy* 6th edition. He is part of pending patent applications to detect and predict seizures and to diagnose epilepsy. He receives research support from the Epilepsy Research Foundation, the American Epilepsy Society, the Epilepsy Foundation of America, the Epilepsy Therapy Project, PCORI, the Pediatric Epilepsy Research Foundation, Cure, HHV-6 Foundation, and received research grants from Lundbeck, Eisai, Upsher-Smith, Acorda, and Pfizer. He serves as a Scientific Advisor for Upsher Smith and Lundbeck. He performs video electroencephalogram long-term and ICU monitoring, electroencephalograms, and other electrophysiological studies at Boston Children's Hospital and affiliated hospitals and bills for these procedures and he evaluates pediatric neurology patients and bills for clinical care. He has received speaker honorariums from national societies including the AAN, AES and ACNS, and for grand rounds at various academic centers. His wife, Dr. Karen Stannard, is a pediatric neurologist and she performs video electroencephalogram long-term and ICU monitoring, electroencephalograms, and other electrophysiological studies and bills for these procedures and she evaluates pediatric neurology patients and bills for clinical care.

Acknowledgement

This manuscript was in part funded by the Epilepsy Research Fund.

References

- [1] Sander JW. The epidemiology of epilepsy revisited. *Curr Opin Neurol* 2003;16(2):165–70.
- [2] Camfield P, Camfield C. Incidence, prevalence and aetiology of seizures and epilepsy in children. *Epileptic Disord* 2015;17(2):117–23.
- [3] Forsgren L. Prevalence of epilepsy in adults in northern Sweden. *Epilepsia* 1992;33(3):450–8.
- [4] Banerjee PN, Filippi D, Allen Hauser W. The descriptive epidemiology of epilepsy – a review. *Epilepsy Res* 2009;85(1):31–45.
- [5] Mac TL, Tran DS, Quet F, Odermatt P, Preux PM, Tan CT. Epidemiology, aetiology, and clinical management of epilepsy in Asia: a systematic review. *Lancet Neurol* 2007;6(6):533–43.
- [6] Kwan P, Brodie MJ. Early identification of refractory epilepsy. *N Engl J Med* 2000;342(5):314–9.
- [7] Forsgren L, Beghi E, Oun A, Sillanpää M. The epidemiology of epilepsy in Europe – a systematic review. *Eur J Neurol* 2005;12(4):245–53.
- [8] Sidenvall R, Forsgren L, Heijbel J. Prevalence and characteristics of epilepsy in children in northern Sweden. *Seizure* 1996;5(2):139–46.
- [9] Christensen J, Pedersen CB, Sidenius P, Olsen J, Vestergaard M. Long-term mortality in refractory partial epilepsy: does surgical treatment make a difference? *J Neurol Neurosurg Psychiatry* 2010;81(7):716–8.
- [10] Holst AG, Winkel BG, Risgaard B, Nielsen JB, Rasmussen PV, Haunsø S, et al. Epilepsy and risk of death and sudden unexpected death in the young: a nationwide study. *Epilepsia* 2013;54(9):1613–20.
- [11] Nashef L, Fish DR, Sander JW, Shorvon SD. Incidence of sudden unexpected death in an adult outpatient cohort with epilepsy at a tertiary referral centre. *J Neurol Neurosurg Psychiatry* 1995;58(4):462–4.
- [12] Sillanpää M, Shinnar S, SUDEP and other causes of mortality in childhood-onset epilepsy. *Epilepsy Behav* 2013;28(2):249–55.
- [13] Loscher W, Schmidt D. Modern antiepileptic drug development has failed to deliver: ways out of the current dilemma. *Epilepsia* 2011;52(4):657–78.
- [14] Neligan A, Bell GS, Shorvon SD, Sander JW. Temporal trends in the mortality of people with epilepsy: a review. *Epilepsia* 2010;51(11):2241–6.
- [15] Jacoby A. Epilepsy and the quality of everyday life. Findings from a study of people with well-controlled epilepsy. *Soc Sci Med* 1992;34(6):657–66.
- [16] Glauser TA, Naan A, Shinnar S, Hirtz DG, Dlugos D, Masur D, et al. Ethosuximide: valproic acid, and lamotrigine in childhood absence epilepsy. *N Engl J Med* 2010;362(9):790–9.
- [17] Glauser TA, Naan A, Shinnar S, Hirtz DG, Dlugos D, Masur D, et al. Ethosuximide, valproic acid, and lamotrigine in childhood absence epilepsy: initial monotherapy outcomes at 12 months. *Epilepsia* 2013;54(1):141–55.
- [18] Nijssen TM, Arends JB, Griep PA, Cluitmans PJ. The potential value of three-dimensional accelerometry for detection of motor seizures in severe epilepsy. *Epilepsy Behav* 2005;7(1):74–84.
- [19] Hoppe C, Poepel A, Elger CE. Epilepsy: accuracy of patient seizure counts. *Arch Neurol* 2007;64(11):1595–9.
- [20] Blum DE, Eskola J, Bortz JJ, Fisher RS. Patient awareness of seizures. *Neurology* 1996;47(1):260–4.
- [21] Kerling F, Mueller S, Pauli E, Stefan H. When do patients forget their seizures? An electroclinical study. *Epilepsy Behav* 2006;9(2):281–5.
- [22] Van de Vel A, Verhaert K, Ceulemans B. Critical evaluation of four different seizure detection systems tested on one patient with focal and generalized tonic and clonic seizures. *Epilepsy Behav* 2014;37:91–4.
- [23] Fisher RS, van Emde Boas W, Blume W, Elger C, Genton P, Lee P, et al. Epileptic seizures and epilepsy: definitions proposed by the International League Against Epilepsy (ILAE) and the International Bureau for Epilepsy (IBE). *Epilepsia* 2005;46(4):470–2.
- [24] Blume WT, Lüders HO, Mizrahi E, Tassinari C, van Emde Boas W, Engel Jr J. Glossary of descriptive terminology for ictal semiology: report of the ILAE task force on classification and terminology. *Epilepsia* 2001;42(9):1212–8.
- [25] Aarabi A, He B. Seizure prediction in intracranial EEG: a patient-specific rule-based approach. *Conf Proc IEEE Eng Med Biol Soc* 2011;2011:2566–9.
- [26] Ozdemir N, Yildirim E. Patient specific seizure prediction system using Hilbert spectrum and Bayesian networks classifiers. *Comput Math Methods Med* 2014;2014:p572082.
- [27] Gotman J. Automatic seizure detection: improvements and evaluation. *Electroencephalogr Clin Neurophysiol* 1990;76(4):317–24.
- [28] Gadhoumi K, Gotman J, Lina JM. Scale invariance properties of intracerebral EEG improve seizure prediction in mesial temporal lobe epilepsy. *PLOS ONE* 2015;10(4):e0121182.
- [29] Park Y, Luo L, Parhi KK, Netoff T. Seizure prediction with spectral power of EEG using cost-sensitive support vector machines. *Epilepsia* 2011;52(10):1761–70.
- [30] Rabbi AF, Azinfar L, Fazel-Rezai R. Seizure prediction using adaptive neuro-fuzzy inference system. *Conf Proc IEEE Eng Med Biol Soc* 2013;2013:2100–3.
- [31] Alexandre Teixeira C, Direito B, Bandarabadi M, Le Van Quyen M, Valderrama M, Schelter B, et al. Epileptic seizure predictors based on computational intelligence techniques: a comparative study with 278 patients. *Comput Methods Programs Biomed* 2014;114(3):324–36.
- [32] Bandarabadi M, Rasekhi J, Teixeira CA, Karami MR, Dourado A. On the proper selection of preictal period for seizure prediction. *Epilepsy Behav* 2015;46:158–66.
- [33] Parvez MZ, Paul M. Epileptic seizure prediction by exploiting spatiotemporal relationship of EEG signals using phase correlation. *IEEE Trans Neural Syst Rehabil Eng* 2015.
- [34] Wang N, Lyu MR. Extracting and selecting distinctive EEG features for efficient epileptic seizure prediction. *IEEE J Biomed Health Inform* 2015;19(5):1648–59.
- [35] Kim H, Rosen J. Epileptic seizure detection – an AR model based algorithm for implantable device. *Conf Proc IEEE Eng Med Biol Soc* 2010;2010:5541–4.
- [36] Petersen EB, Duun-Henriksen J, Mazzaretto A, Kjær TW, Thomsen CE, Sørensen HB. Generic single-channel detection of absence seizures. *Conf Proc IEEE Eng Med Biol Soc* 2011;2011:4820–3.
- [37] Luca S, Karsmakers P, Cuppens K, Croonenborghs T, Van de Vel A, Ceulemans B, et al. Detecting rare events using extreme value statistics applied to epileptic convulsions in children. *Artif Intell Med* 2014;60(2):89–96.
- [38] Zandi AS, Dumont GA, Javidan M, Tafreshi R. Epileptic seizure prediction using variational mixture of Gaussians. *Conf Proc IEEE Eng Med Biol Soc* 2011;2011:7549–52.
- [39] Bandarabadi M, Teixeira CA, Rasekhi J, Dourado A. Epileptic seizure prediction using relative spectral power features. *Clin Neurophysiol* 2015;126(2):237–48.
- [40] Duun-Henriksen J, Madsen RE, Remvig LS, Thomsen CE, Sørensen HB, Kjær TW. Automatic detection of childhood absence epilepsy seizures: toward a monitoring device. *Pediatr Neurol* 2012;46(5):287–92.
- [41] Schulze-Bonhage A, Sales F, Wagner K, Teotonio R, Carius A, Schelle A, et al. Views of patients with epilepsy on seizure prediction devices. *Epilepsy Behav* 2010;18(4):388–96.

- [43] Waterhouse E. New horizons in ambulatory electroencephalography. *IEEE Eng Med Biol Mag* 2003;22(3):74–80.
- [44] Luan B, Sun M. A simulation study on a single-unit wireless EEG sensor. *Proc IEEE Annu Northeast Bioeng Conf* 2015;2015.
- [45] Luan B, Jia W, Thirumala PD, Balzer J, Gao D, Sun M. A feasibility study on a single-unit wireless EEG sensor. *Int Conf Signal Process Proc* 2014;2014:2282–5.
- [46] Wyckoff SN, Sherlin LH, Ford NL, Dalke D. Validation of a wireless dry electrode system for electroencephalography. *J Neuroeng Rehabil* 2015;12(1):95.
- [47] Do Valle BG, Cash SS, Sodini CG. Wireless behind-the-ear EEG recording device with wireless interface to a mobile device (iPhone/iPod touch). *Conf Proc IEEE Eng Med Biol Soc* 2014;2014:5952–5.
- [48] Mihajlovic V, Grundlehner B, Vullers R, Penders J. Wearable, wireless EEG solutions in daily life applications: what are we missing? *IEEE J Biomed Health Inform* 2015;19(1):6–21.
- [49] Grant AC, Abdel-Baki SG, Omurtag A, Sinert R, Chari G, Malhotra S, et al. Diagnostic accuracy of microEEG: a miniature, wireless EEG device. *Epilepsy Behav* 2014;34:81–5.
- [50] Grummett TS, Leibbrandt RE, Lewis TW, DeLosAngeles D, Powers DM, Wiloughby JO, et al. Measurement of neural signals from inexpensive: wireless and dry EEG systems. *Physiol Meas* 2015;36(7):1469–84.
- [51] David Hairston W, Whitaker KW, Ries AJ, Vettel JM, Courtney Bradford J, Kerick SE, et al. Usability of four commercially-oriented EEG systems. *J Neural Eng* 2014;11(4):046018.
- [52] Jakab A, Kulkas A, Salpavaara T, Kauppinen P, Verho J, Heikkilä H, et al. Novel wireless electroencephalography system with a minimal preparation time for use in emergencies and prehospital care. *Biomed Eng Online* 2014;13:60.
- [53] Lehmkuhle M, Elwood M, Wheeler J, Fisher J, Dudek FE. Development of a discrete, wearable, EEG device for counting seizures (abstract). In: 69th Annual Meeting of the American Epilepsy Society. AES; 2015. Abstract nr 2.158, 2015.
- [54] Larsen SN, Conradsen I, Beniczky S, Sorensen HB. Detection of tonic epileptic seizures based on surface electromyography. *Conf Proc IEEE Eng Med Biol Soc* 2014;2014:942–5.
- [55] Conradsen I, Beniczky S, Hoppe K, Wolf P, Sorensen HB. Automated algorithm for generalized tonic-clonic epileptic seizure onset detection based on sEMG zero-crossing rate. *IEEE Trans Biomed Eng* 2012;59(2):579–85.
- [56] Conradsen I, Wolf P, Sams T, Sorensen HB, Beniczky S. Patterns of muscle activation during generalized tonic and tonic-clonic epileptic seizures. *Epilepsia* 2011;52(11):2125–32.
- [57] Conradsen I, Beniczky S, Wolf P, Jennum P, Sorensen HB. Evaluation of novel algorithm embedded in a wearable sEMG device for seizure detection. *Conf Proc IEEE Eng Med Biol Soc* 2012;2012:2048–51.
- [58] Szabó C, Morgan LC, Karkar KM, Leary LD, Lie OV, Girouard M, et al. Electromyography-based seizure detection: Preliminary results comparing a generalized tonic-clonic seizure detection algorithm to video-EEG recordings. *Epilepsia* 2015.
- [59] Poh MZ, Swenson NC, Picard RW. A wearable sensor for unobtrusive: long-term assessment of electrodermal activity. *IEEE Trans Biomed Eng* 2010;57(5):1243–52.
- [60] Poh MZ, Loddenkemper T, Swenson NC, Goyal S, Madsen JR, Picard RW. Continuous monitoring of electrodermal activity during epileptic seizures using a wearable sensor. *Conf Proc IEEE Eng Med Biol Soc* 2010;2010:4415–8.
- [61] Poh MZ, Loddenkemper T, Reinsberger C, Swenson NC, Goyal S, Madsen JR, et al. Autonomic changes with seizures correlate with postictal EEG suppression. *Neurology* 2012;78(23):1868–76.
- [62] Sarkis RA, Thome-Souza S, Poh MZ, Llewellyn N, Klehm J, Madsen JR, et al. Autonomic changes following generalized tonic clonic seizures: an analysis of adult and pediatric patients with epilepsy. *Epilepsy Res* 2015;115:113–8.
- [63] Nickels KC, Grossardt BR, Wirrell EC. Epilepsy-related mortality is low in children: a 30-year population-based study in Olmsted County, MN. *Epilepsia* 2012;53(12):2164–71.
- [64] van Elmpy WJ, Nijssen TM, Griep PA, Arends JB. A model of heart rate changes to detect seizures in severe epilepsy. *Seizure* 2006;15(6):366–75.
- [65] Opherk C, Coromilas J, Hirsch IJ. Heart rate and EKG changes in 102 seizures: analysis of influencing factors. *Epilepsy Res* 2002;52(2):117–27.
- [66] Behbahani S, Dabanloo NJ, Nasrabadi AM, Teixeira CA, Dourado A. Pre-ictal heart rate variability assessment of epileptic seizures by means of linear and non-linear analyses. *Anadolu Kardiyol Derg* 2013;13(8):797–803.
- [67] Osorio I, Manly BF. Is seizure detection based on EKG clinically relevant? *Clin Neurophysiol* 2014;125(10):1946–51.
- [68] Zijlmans M, Flanagan D, Gotman J. Heart rate changes and ECG abnormalities during epileptic seizures: prevalence and definition of an objective clinical sign. *Epilepsia* 2002;43(8):847–54.
- [69] Jeppesen J, Beniczky S, Johansen P, Sidenius P, Fuglsang-Frederiksen A. Detection of epileptic seizures with a modified heart rate variability algorithm based on Lorenz plot. *Seizure* 2015;24:1–7.
- [70] van Andel J, Ungureanu C, Aarts R, Leijten F, Arends J. Using photoplethysmography in heart rate monitoring of patients with epilepsy. *Epilepsy Behav* 2015;45:142–5.
- [71] Kerem DH, Geva AB. Forecasting epilepsy from the heart rate signal. *Med Biol Eng Comput* 2005;43(2):230–9.
- [72] Sei H, Furuno N, Morita Y. Diurnal changes of blood pressure: heart rate and body temperature during sleep in the rat. *J Sleep Res* 1997;6(2):113–9.
- [73] Mancina G, Ferrari A, Gregorini L, Parati G, Pomidossi G, Bertinieri G, et al. Blood pressure and heart rate variabilities in normotensive and hypertensive human beings. *Circ Res* 1983;53(1):96–104.
- [74] Leutmezer F, Scherthauer C, Lurger S, Pötzelberger K, Baumgartner C. Electrocardiographic changes at the onset of epileptic seizures. *Epilepsia* 2003;44(3):348–54.
- [75] Jansen K, Varon C, Van Huffel S, Lagae L. Peri-ictal ECG changes in childhood epilepsy: implications for detection systems. *Epilepsy Behav* 2013;29(1):72–6.
- [76] Nilsen KB, Haram M, Tangedal S, Sand T, Brodtkorb E. Is elevated pre-ictal heart rate associated with secondary generalization in partial epilepsy? *Seizure* 2010;19(5):291–5.
- [77] Osorio I. Automated seizure detection using EKG. *Int J Neural Syst* 2014;24(2):1450001.
- [78] Massé F, van Bussel M, Serteyn A, Arends J, Penders J. Miniaturized wireless ECG monitor for real-time detection of epileptic seizures. *ACM Trans Embed Comput Syst* 2010;12(4):1–21.
- [79] Romero I, Grundlehner B, Penders J. Robust beat detector for ambulatory cardiac monitoring. *Conf Proc IEEE Eng Med Biol Soc* 2009;2009:950–3.
- [80] Kwon S, Kim H, Park KS. Validation of heart rate extraction using video imaging on a built-in camera system of a smartphone. *Conf Proc IEEE Eng Med Biol Soc* 2012;2012:2174–7.
- [81] Boon P, Vonck K, van Rijckevorsel K, El Tahry R, Elger CE, Mullatti N, et al. A prospective, multicenter study of cardiac-based seizure detection to activate vagus nerve stimulation. *Seizure* 2015;32:52–61.
- [82] Ramgopal S, Thome-Souza S, Jackson M, Kadish NE, Sánchez Fernández I, Klehm J, et al. Seizure detection, seizure prediction, and closed-loop warning systems in epilepsy. *Epilepsy Behav* 2014;37:291–307.
- [83] Beniczky S, Polster T, Kjaer TW, Hjalgrim H. Detection of generalized tonic-clonic seizures by a wireless wrist accelerometer: a prospective, multicenter study. *Epilepsia* 2013;54(4):e58–61.
- [84] Lockman J, Fisher RS, Olson DM. Detection of seizure-like movements using a wrist accelerometer. *Epilepsy Behav* 2011;20(4):638–41.
- [85] Nijssen TM, Aarts RM, Cluitmans PJ, Griep PA. Time-frequency analysis of accelerometer data for detection of myoclonic seizures. *IEEE Trans Inf Technol Biomed* 2010;14(5):1197–203.
- [86] Van de Vel A, Cuppens K, Bonroy B, Milosevic M, Van Huffel S, Vanrumste B, et al. Long-term home monitoring of hypermotor seizures by patient-worn accelerometers. *Epilepsy Behav* 2013;26(1):118–25.
- [87] Kramer U, Kipervasser S, Shlitzer A, Kuzniecky R. A novel portable seizure detection alarm system: preliminary results. *J Clin Neurophysiol* 2011;28(1):36–8.
- [88] Nijssen TM, Aarts RM, Arends JB, Cluitmans PJ. Model for arm movements during myoclonic seizures. *Conf Proc IEEE Eng Med Biol Soc* 2007;2007:1582–5.
- [89] Dalton A, Patel S, Chowdhury AR, Welsh M, Pang T, Schachter S, et al. Development of a body sensor network to detect motor patterns of epileptic seizures. *IEEE Trans Biomed Eng* 2012;59(11):3204–11.
- [90] Schulc E, Unterberger I, Saboor S, Hilbe J, Ertl M, Ammenwerth E, et al. Measurement and quantification of generalized tonic-clonic seizures in epilepsy patients by means of accelerometry—an explorative study. *Epilepsy Res* 2011;95(1–2):173–83.
- [91] Patterson AL, Mudigoudar B, Fulton S, McGregor A, Poppel KV, Wheless MC, et al. SmartWatch by SmartMonitor: assessment of seizure detection efficacy for various seizure types in children, a large prospective single-center study. *Pediatr Neurol* 2015;53(4):309–11.
- [92] Jallon P, Bonnet S, Antonakios M, Guillemaud R. Detection system of motor epileptic seizures through motion analysis with 3D accelerometers. *Conf Proc IEEE Eng Med Biol Soc* 2009;2009:2466–9.
- [93] Pediaditis M, Tsiknakis M, Leitgeb N. Vision-based motion detection: analysis and recognition of epileptic seizures – a systematic review. *Comput Methods Programs Biomed* 2012;108(3):1133–48.
- [94] Lu H, Pan Y, Mandal B, Eng HL, Guan C, Chan DW. Quantifying limb movements in epileptic seizures through color-based video analysis. *IEEE Trans Biomed Eng* 2013;60(2):461–9.
- [95] Cuppens K, Chen CW, Wong KB, Van de Vel A, Lagae L, Ceulemans B, et al. Using spatio-temporal interest points (STIP) for myoclonic jerk detection in nocturnal video. *Conf Proc IEEE Eng Med Biol Soc* 2012;2012:4454–7.
- [96] Kalitzin S, Petkov G, Velis D, Vledder B, Lopes da Silva F. Automatic segmentation of episodes containing epileptic clonic seizures in video sequences. *IEEE Trans Biomed Eng* 2012;59(12):3379–85.
- [97] Mandal B, Eng HL, Lu H, Chan DW, Ng YL. Non-intrusive head movement analysis of videotaped seizures of epileptic origin. *Conf Proc IEEE Eng Med Biol Soc* 2012;2012:6060–3.
- [98] Rémi J, Cunha JP, Vollmar C, Topcuoğlu ÖB, Meier A, Ulowetz S, et al. Quantitative movement analysis differentiates focal seizures characterized by automatisms. *Epilepsy Behav* 2011;20(4):642–7.
- [99] Rémi J, Wagner P, O'Dwyer R, Silva Cunha JP, Vollmar C, Krotofil I, et al. Ictal head turning in frontal and temporal lobe epilepsy. *Epilepsia* 2011;52(8):1447–51.
- [100] O'Dwyer R, Silva Cunha JP, Vollmar C, Mauerec C, Feddersen B, Burgess RC, et al. Lateralizing significance of quantitative analysis of head movements before secondary generalization of seizures of patients with temporal lobe epilepsy. *Epilepsia* 2007;48(3):524–30.

- [101] Lamberts RJ, Thijs RD, Laffan A, Langan Y, Sander JW. Sudden unexpected death in epilepsy: people with nocturnal seizures may be at highest risk. *Epilepsia* 2012;53(2):253–7.
- [102] Carlson C, Arnedo V, Cahill M, Devinsky O. Detecting nocturnal convulsions: efficacy of the MP5 monitor. *Seizure* 2009;18(3):225–7.
- [103] Fulton S, Poppel KV, McGregor A, Ellis M, Patters A, Wheless J. Prospective study of 2 bed alarms for detection of nocturnal seizures. *J Child Neurol* 2012;28(11):1430–3.
- [104] Langan Y, Nashef L, Sander JW. Case-control study of SUDEP. *Neurology* 2005;64(7):1131–3.
- [105] Poppel KV, Fulton SP, McGregor A, Ellis M, Patters A, Wheless J. Prospective study of the Emfit movement monitor. *J Child Neurol* 2013;28(11):1434–6.
- [106] Brown SW, Goldstein LH. Can seizure-alert dogs predict seizures? *Epilepsy Res* 2011;97(3):236–42.
- [107] Kirtan A, Wirrell E, Zhang J, Hamiwka L. Seizure-alerting and -response behaviors in dogs living with epileptic children. *Neurology* 2004;62(12):2303–5.
- [108] Dalziel DJ, Uthman BM, McGorray SP, Reep RL. Seizure-alert dogs: a review and preliminary study. *Seizure* 2003;12(2):115–20.
- [109] Strong V, Brown S, Huyton M, Coyle H. Effect of trained seizure alert dogs on frequency of tonic-clonic seizures. *Seizure* 2002;11(6):402–5.
- [110] Strong V, Brown SW, Walker R. Seizure-alert dogs – fact or fiction? *Seizure* 1999;8(1):62–5.
- [111] Ortiz R, Liporace J. Seizure-alert dogs: observations from an inpatient video/EEG unit. *Epilepsy Behav* 2005;6(4):620–2.
- [112] Doherty MJ, Haltiner AM. Wag the dog: skepticism on seizure alert canines. *Neurology* 2007;68(4):p309.
- [113] Krauss GL, Choi JS, Lesser RP. Pseudoseizure dogs. *Neurology* 2007;68(4):308–9.
- [114] Weinand ME, Carter LP, Patton DD, Oommen KJ, Labiner DM, Talwar D. Long-term surface cortical cerebral blood flow monitoring in temporal lobe epilepsy. *Neurosurgery* 1994;35(4):657–64.
- [115] Federico P, Abbott DF, Briellmann RS, Harvey AS, Jackson GD. Functional MRI of the pre-ictal state. *Brain* 2005;128(Pt 8):1811–7.
- [116] Moseley BD, Britton JW, So E. Increased cerebral oxygenation precedes generalized tonic clonic seizures. *Epilepsy Res* 2014;108(9):1671–4.
- [117] Seyal M. Frontal hemodynamic changes precede EEG onset of temporal lobe seizures. *Clin Neurophysiol* 2014;125(3):442–8.
- [118] Roche-Labarbe N, Zaaïmi B, Berquin P, Nehlig A, Grebe R, Wallois F. NIRS-measured oxy- and deoxyhemoglobin changes associated with EEG spike-and-wave discharges in children. *Epilepsia* 2008;49(11):1871–80.
- [119] Jeppesen J, Beniczky S, Johansen P, Sidenius P, Fuglsang-Frederiksen A. Exploring the capability of wireless near infrared spectroscopy as a portable seizure detection device for epilepsy patients. *Seizure* 2015;26:43–8.
- [120] Cook MJ, O'Brien TJ, Berkovic SF, Murphy M, Morokoff A, Fabinyi G, et al. Prediction of seizure likelihood with a long-term, implanted seizure advisory system in patients with drug-resistant epilepsy: a first-in-man study. *Lancet Neurol* 2013;12(6):563–71.
- [121] Kim B, Nogueira AB, Thome-Souza S, Kapur K, Klehm J, Jackson M, et al. Diurnal and nocturnal patterns of autonomic neurophysiological measurements are related to timing of seizures. In: Poster presented at: CNS 2015. Child Neurology Society annual meeting; 2015.
- [122] Singh K, Katz ES, Zarowski M, Loddenkemper T, Llewellyn N, Manganaro S, et al. Cardiopulmonary complications during pediatric seizures: a prelude to understanding SUDEP. *Epilepsia* 2013;54(6):1083–91.
- [123] Johnston SC, Horn JK, Valente J, Simon RP. The role of hypoventilation in a sheep model of epileptic sudden death. *Ann Neurol* 1995;37(4):531–7.
- [124] Bateman LM, Spitz M, Seyal M. Ictal hypoventilation contributes to cardiac arrhythmia and SUDEP: report on two deaths in video-EEG-monitored patients. *Epilepsia* 2010;51(5):916–20.
- [125] Conradsen I, Beniczky S, Wolf P, Kjaer TW, Sams T, Sorensen HB. Automatic multi-modal intelligent seizure acquisition (MISA) system for detection of motor seizures from electromyographic data and motion data. *Comput Methods Programs Biomed* 2012;107(2):97–110.
- [126] Milosevic M, Van de Vel A, Bonroy B, Ceulemans B, Lagae L, VanRumste B, et al. Automated detection of tonic-clonic seizures using 3D accelerometry and surface electromyography in pediatric patients. *IEEE J Biomed Health Inform* 2015.
- [127] Conradsen I, Beniczky S, Wolf P, Terney D, Sams T, Sorensen HB. Multi-modal intelligent seizure acquisition (MISA) system – a new approach towards seizure detection based on full body motion measures. *Conf Proc IEEE Eng Med Biol Soc* 2009;2009:2591–5.
- [128] Conradsen I, Beniczky S, Wolf P, Henriksen J, Sams T, Sorensen HB. Seizure onset detection based on a uni- or multi-modal intelligent seizure acquisition (UISA/MISA) system. *Conf Proc IEEE Eng Med Biol Soc* 2010;2010:3269–72.
- [129] Poh MZ, Loddenkemper T, Reinsberger C, Swenson NC, Goyal S, Sabtala MC, et al. Convulsive seizure detection using a wrist-worn electrodermal activity and accelerometry biosensor. *Epilepsia* 2012;53(5):e93–7.
- [130] Becq G, Bonnet S, Minotti L, Antonakios M, Guillemaud R, Kahane P. Collection and exploratory analysis of attitude sensor data in an epilepsy monitoring unit. *Conf Proc IEEE Eng Med Biol Soc* 2007;2007:2775–8.
- [131] Bonnet S, Heliot R. A magnetometer-based approach for studying human movements. *IEEE Trans Biomed Eng* 2007;54(7):1353–5.
- [132] Becq G, Kahane P, Minotti L, Bonnet S, Guillemaud R. Classification of epileptic motor manifestations and detection of tonic-clonic seizures with acceleration norm entropy. *IEEE Trans Biomed Eng* 2013;60(8):2080–8.
- [133] Shoeb A, Pang T, Gutttag J, Schachter S. Non-invasive computerized system for automatically initiating vagus nerve stimulation following patient-specific detection of seizures or epileptiform discharges. *Int J Neural Syst* 2009;19(3):157–72.
- [134] Shoeb A, Bourgeois B, Treves S, Schachter SC, Gutttag J. Impact of patient-specificity on seizure onset detection performance. *Conf Proc IEEE Eng Med Biol Soc* 2007;2007:4110–4.
- [135] Hoppe C, Feldmann M, Blachut B, Surges R, Elger CE, Helmstaedter C. Novel techniques for automated seizure registration: patients' wants and needs. *Epilepsy Behav* 2015;52(Pt A):1–7.
- [136] Casson A, Yates D, Smith S, Duncan J, Rodriguez-Villegas E. Wearable electroencephalography. What is it, why is it needed, and what does it entail? *IEEE Eng Med Biol Mag* 2010;29(3):44–56.
- [137] Loddenkemper T, Pan A, Neme S, Baker KB, Rezaei AR, Dinner DS, et al. Deep brain stimulation in epilepsy. *J Clin Neurophysiol* 2001;18(6):514–32.
- [138] Alexopoulos AV, Gonugunta V, Yang J, Boulis NM. Electrical stimulation and gene-based neuromodulation for control of medically-refractory epilepsy. *Acta Neurochir Suppl* 2007;97(Pt 2):293–309.
- [139] Velasco M, Velasco F, Velasco AL. Centromedian-thalamic and hippocampal electrical stimulation for the control of intractable epileptic seizures. *J Clin Neurophysiol* 2001;18(6):495–513.